Introduction to Python

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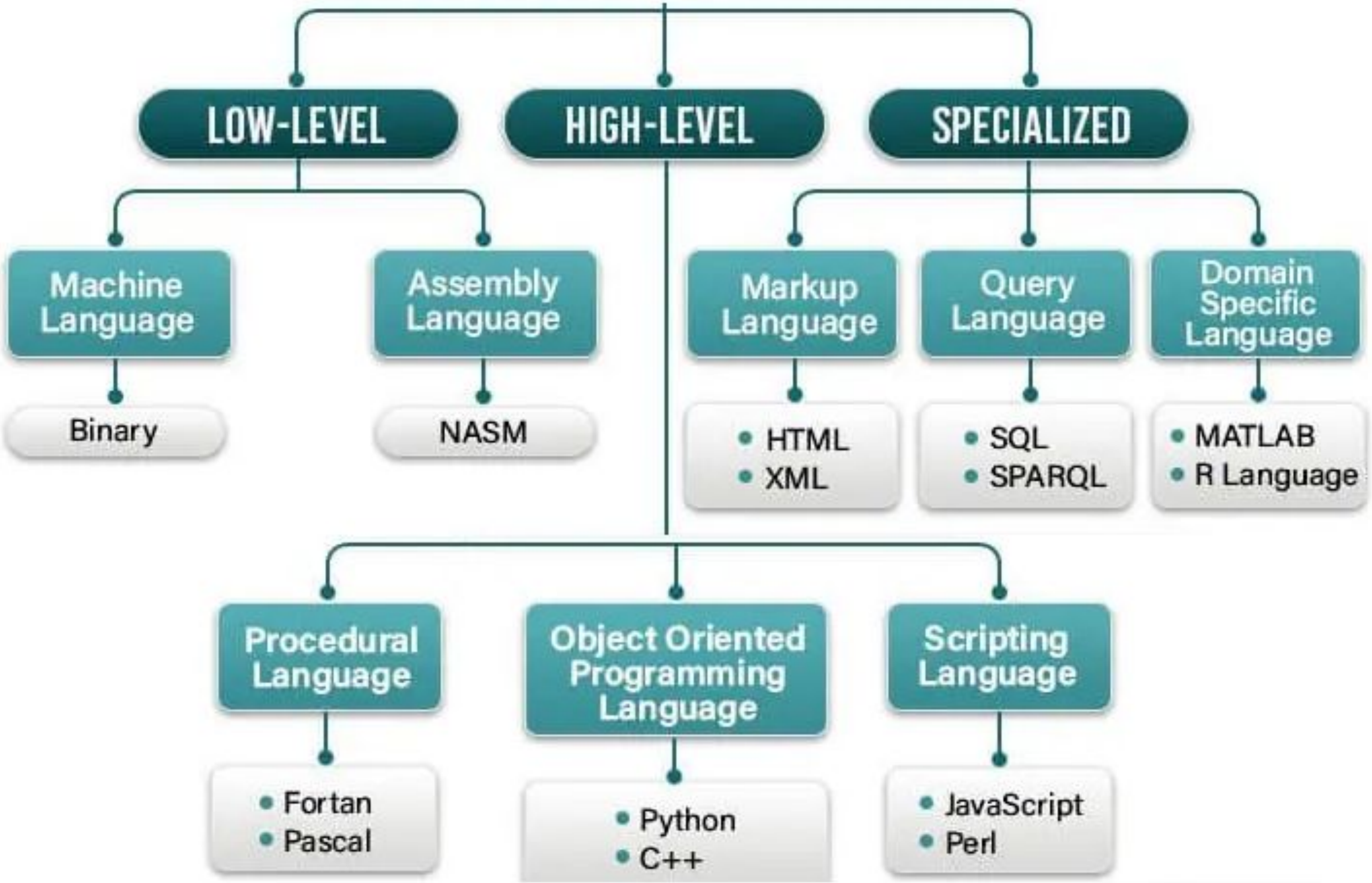
Types of Computer Languages



Types of Programming Languages

Machine Language

1. Directly run on CPU
2. Series of bits like, 0s and 1s
3. Tedious and error prone to write code manually.
4. Not portable

Assembly Language

1. Less error-prone
2. Coding easier than machine Language
3. Replaces 1 and 0s with English instructions
4. Mnemonic codes for corresponding machine language

```
Mov A1, AA
JMP L20
CMP R0, R1
ADD R1, AH, BH
```

MIPS

nasm

x86

High Level Language

1. It is portable
2. Statements are like English language
3. Amount of abstraction provided defines level of programming language

```
void main()
{
  int a=25;
  a = a + 25;
  printf("Hello");
}
```

Java

python

Procedural

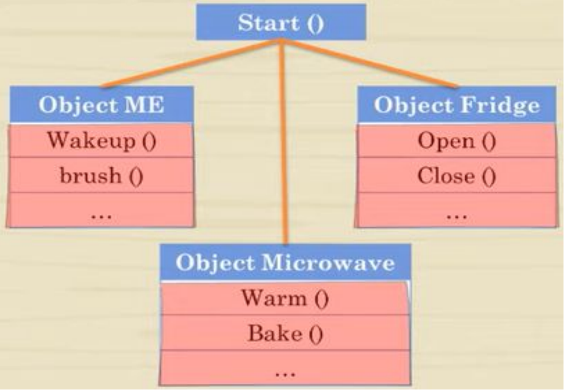
Program is written as sequence of instructions

e.g.: Recipes,
Morning Steps:

Start ()
I wake up
I brush my teeth
I open fridge
I take milk
I warm it in microwave

Object Oriented

Program is an interaction of functions between objects



Procedural	Object Oriented
Program is written as sequence of instructions	Program is an interaction of functions between objects
1. Top down approach	1. Bottom up approach
2. It doesn't have proper way for hiding data	2. Helps in wrapping data and functions in a class
3. Data is not secure	3. Helps Building secure programs
4. Code is interdependent	4. Code is modular
5. Reuse difficult	5. Can be extended for reuse

Compiler Vs Interpreter

High-level programming language into machine code

A compiler and an interpreter are both used to translate code written in a high-level programming language into machine code, but they differ in the way they do so

Compiler

Translates the entire source code into machine code before execution, which makes the program faster to run. A compiler saves the translated code as a file.

Examples : C, C++, COBOL, and Fortran

Interpreter

Translates the code line by line as the program runs, which makes it easier to detect errors. However, this process can be slower than a compiler. Interpreters don't save the translated code.

Examples : **Python**, JavaScript, Perl, and BASIC

Intermediate Language / code

During the translation of a source program into the machine code for a target machine, a compiler may generate a middle-level language code. It is interpreted by a virtual machine (like Java's bytecode in the JVM) into machine code.

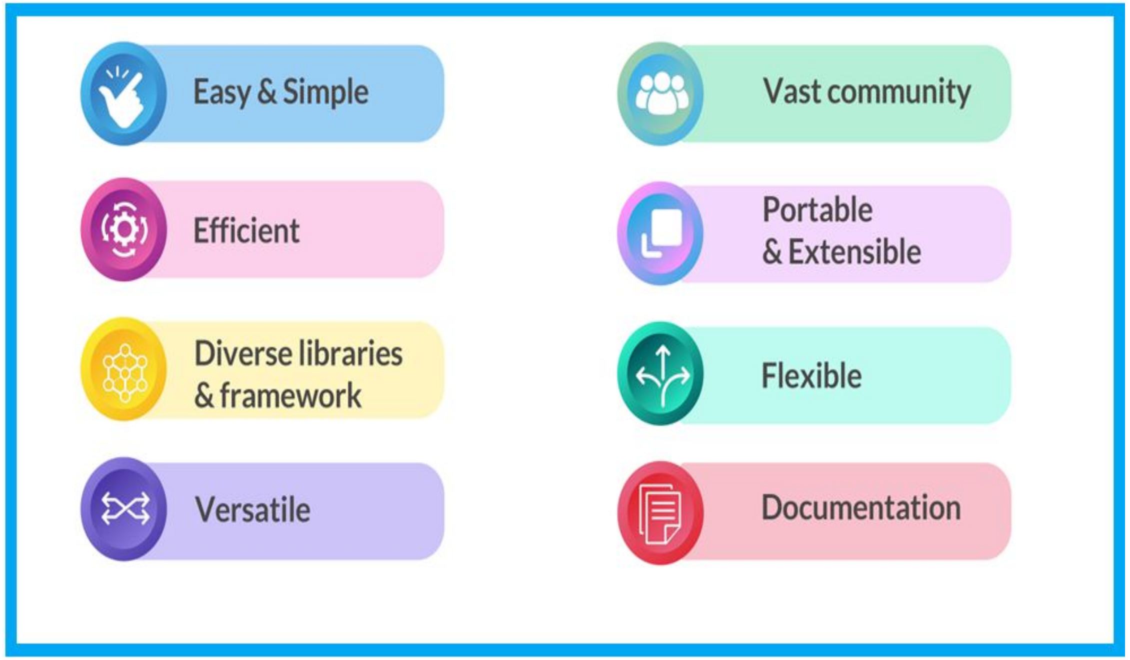
Example: **Java, .NET**

Why Python ?

As of September 2024, as per PYPL (**Popularity** of **P**rogramming **L**anguage Index) – Python language tutorials are most searched on Google

https://pypl.github.io/PYPL.html#google_vignette

Features of Python



Applications of Python

History of Python

- **Python's Early Development**:
  - Began in1980s, with its implementation starting at the end of 1989 by **Guido van Rossum**, who led its development until 2018.
- **Python 1.0 Release**:
  - Released in January 1994, with key features like lambda, reduce, map, and filter.
- **Python 2.0 Release**:
  - Launched in 2000, introducing new features like the garbage collector.
- **Python 3.0 ("Python 3000" or "Py3K")**:

  Released in December 2008, it made significant changes like making print a built-in function and removing backward-compatibility features.
- **Python implementations**
  - **CPython:**

    The original Python implementation, written in C. It's considered the "standard" Python.
  - **Jython:**

    Written in Java, allowing Python code to run on the Java Virtual Machine (JVM).
  - **IronPython:**

    Written in C# for the .NET framework.
  - **MicroPython**

    Implementation of Python for microcontrollers and other constrained hardware.

Thank you